Power 1V8-3V3-5V

File: Power.kicad_sch

Controller

File: MCU-Sensors.kicad_sch

Driver

File: Driver.kicad_sch

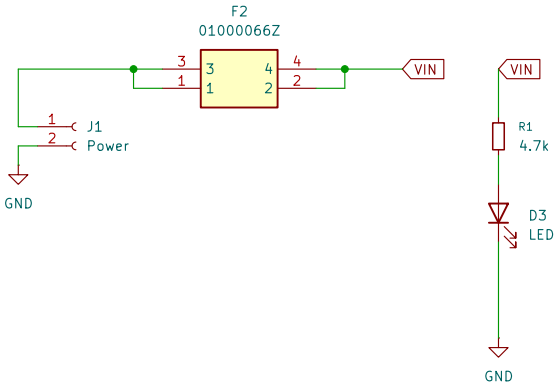
This sheet contains 3 different hierarchical sheets

Sheet: /
File: Mobile_controller.kicad_sch

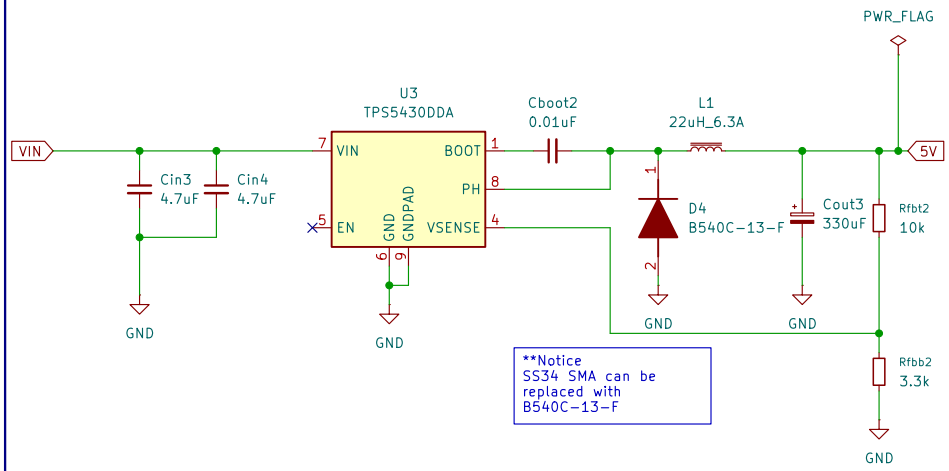
Title: Main sheet

Size: A5	Date: 2026-08-27	**Rev:**
KiCad E.D.A. 10.0.2		Id: 1/4

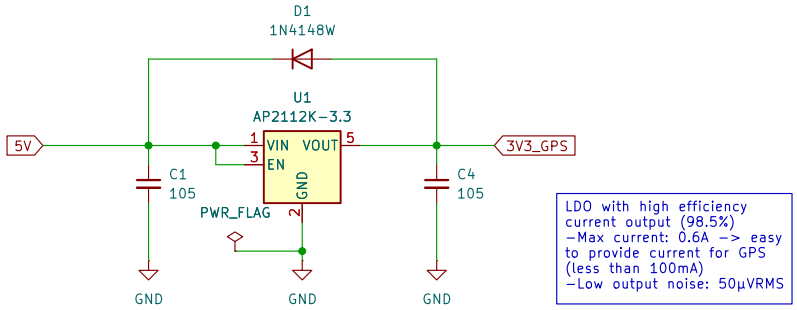
LED



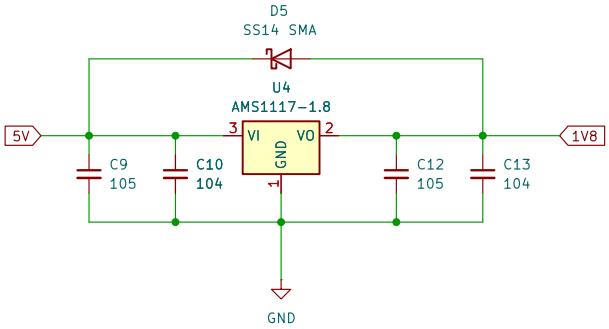
Step down converter



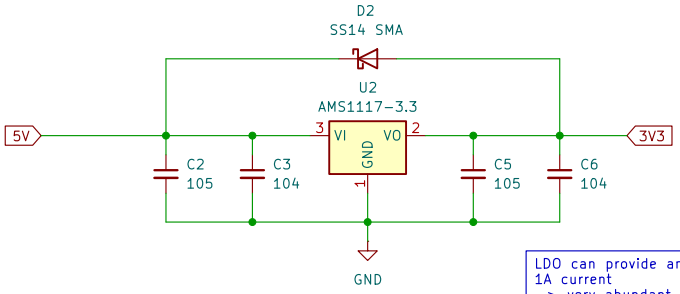
LDO 3.3V for GPS



LDO 1.8V (for IMU)



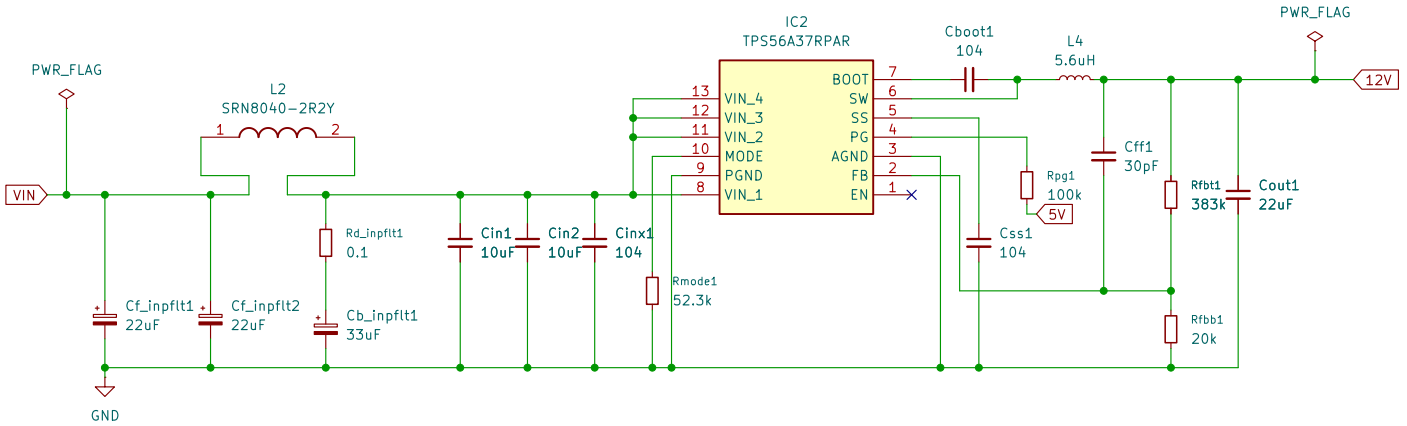
LDO 3.3V (Except for GPS)



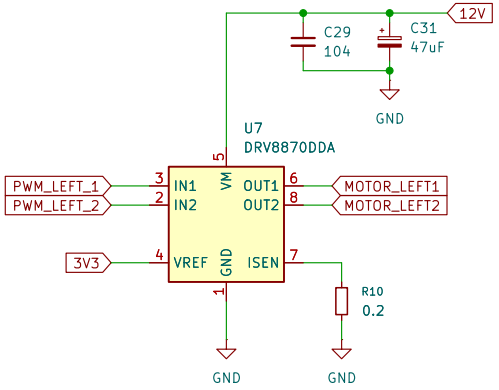
Diodes are used to protect from reverse polarity and discharge
Connector J3 use to connect power source from driver to controller

Sheet: /Power 1V8-3V3-5V/		
File: Power.kicad_sch		
Title: Power		
Size: A4	Date: 2026-08-27	**Rev:**
KiCad E.D.A. 10.0.2		Id: 2/4

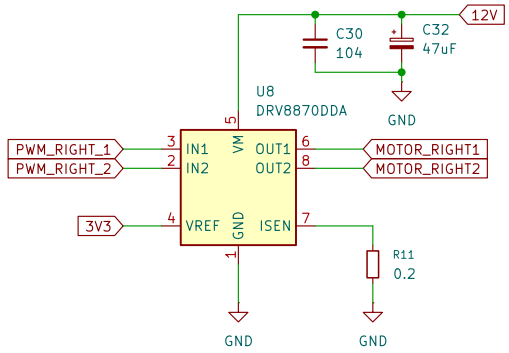
DC SOURCE FOR LEFT MOTOR



LEFT DRIVER



RIGHT DRIVER







H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

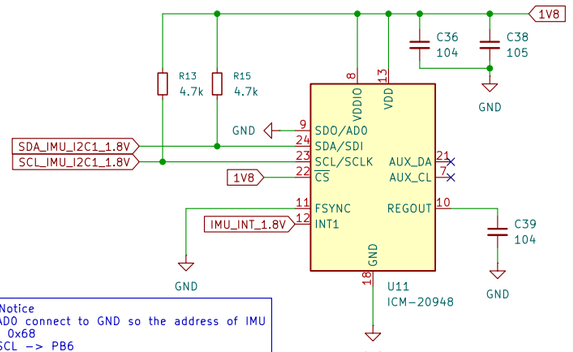
GPIOs having PWM need to be on the same timer
Driver's datasheet implies that control speed through PWM between "Drive" and "Brake"

Sheet: /Driver/
File: Driver.kicad_sch

Title: Driver

Size: A4	Date: 2026-08-27	Rev:
KiCad E.D.A. 10.0.2		Id: 3/4

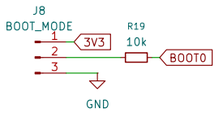
IMU



**Notice
-ADO connect to GND so the address of IMU is: 0x68
-SCL -> PB6
-SDA -> PB7
-FSYNC is frame synchronization using for synchronize frame in camera vibration context (connect to GND to disable it)
-AUX_DA and AUX_CL is I2C bus for external sensors

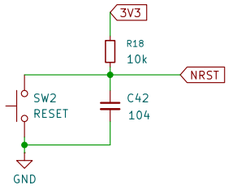
BOOT States Switch

**Notice
-BOOT0 = 1 (3V3) is system memory boot (USB - STM32CubeProgrammer)
-BOOT0 = 0 (GND) is main flash memory boot (STlink V2- STM32CubeIDE)
-Resistor 10k to limit the current to BOOT0 when switching from 3V3 to GND and prevent floating when flip the switch (using internal stray capacitor and resistor to create a LPF) to hold the logic voltage
Use jumper to switch BOOT mode

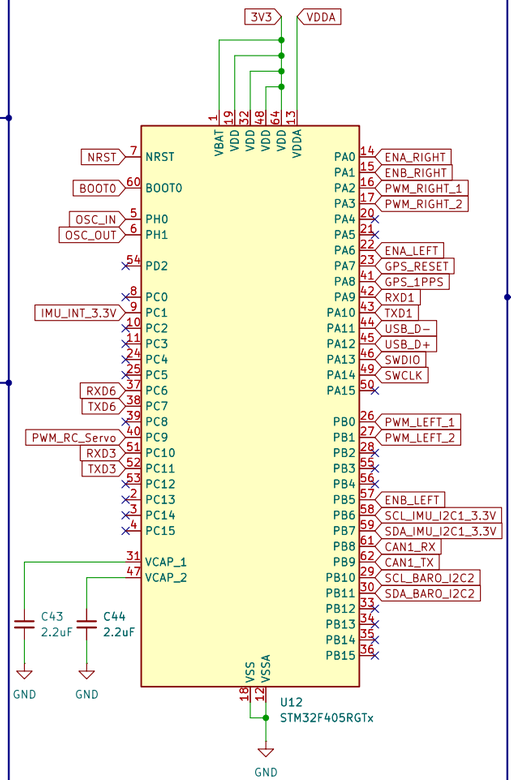


Reset button

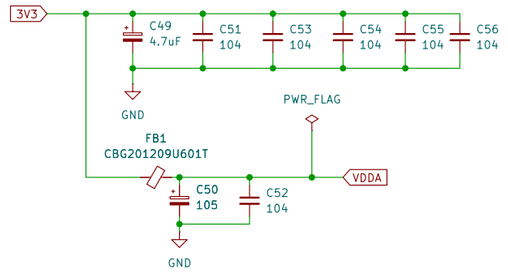
**Notice
-Pull up resistor to prevent microcontroller rebooting unexpectedly from noise
-Capacitor also prevent noise and pull noise down to GND and also help delay
since capacitor need to recharge -> stablize the board when reset



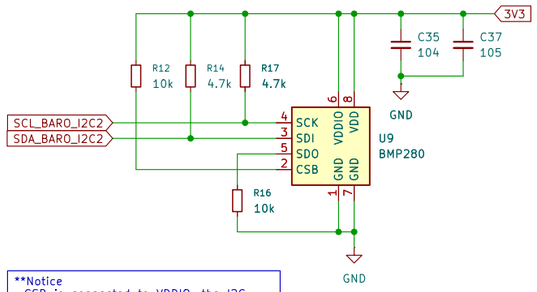
MCU



Decoupling MCU

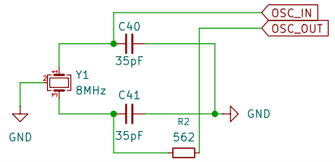


Barometer



**Notice
-CSB is connected to VDDIO, the I2C interface is active (SPI if connect to GND)
-SDO pin is connected to GND so the address will be 0x76
-SDO pin cannot be left floating or the I2C address will be undefined

Crystal Oscillator



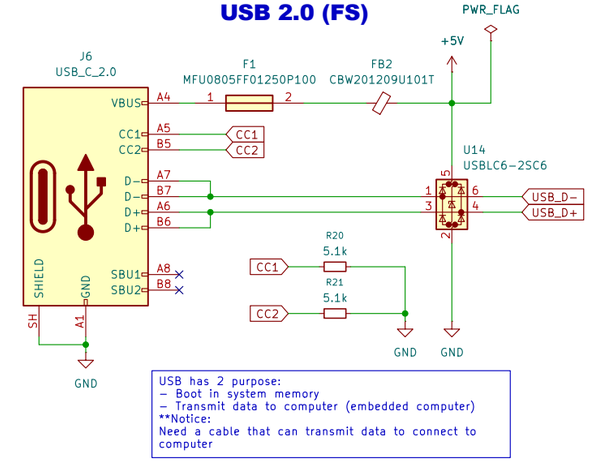
**Notice
Crystal load capacitors formula:
CL = CStray = (C1 * C2) / (C1 + C2)
*Parameters:
-CL: Load capacitance of crystal (20pF)
-CStray: Stray capacitance (2-5pF)

External resistor formula:
REXt = 1 / (2 π F CL2)

*Parameters:
-REXt: External resistor
-F: oscillation frequency of crystal
-CL2: value of the capacitors in PH1 line
**Notice:
Gain margin needs to bigger than 5 to start the oscilator
Gain margin = g_m/g_mcrit_new
-g_m: oscillator transconductance(độ dẫn truyền dao động) (4-26Mhz)
-g_mcrit_new: minimal transconductance to maintain a stable oscillation
g_mcrit_new = 4 * (ESR + REXt) * (2*π*F) * ((C1 + C2)^2)

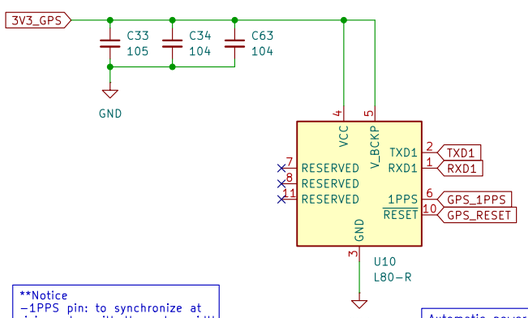
*Parameters
ESR: Equivalent series resistance in oscillator datasheet

USB 2.0 (FS)



USB has 2 purpose:
- Boot in system memory
- Transmit data to computer (embedded computer)
**Notice:
Need a cable that can transmit data to connect to computer

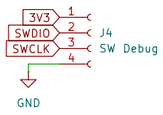
GPS



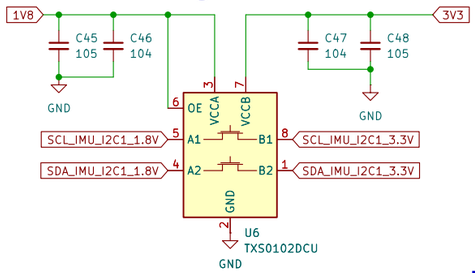
**Notice
-1PPS pin: to synchronize at rising edge with the pulse width 100ms
->This pulse can be used to determine when to collect data from GPS
-RESET pin: reset GPS actively when appears trouble

Automatic power switching using Dual diode schottky (BAT54C)
*Principle:
-If the main source VCC is operating
-> diode will use it to provide V_BCKP
-Else use coin cell battery to keep it operate at minimum current

SW DEBUG

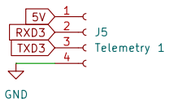


Logic shifter

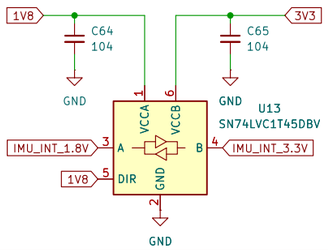


Logic shifter to convert 1.8V I2C signal to 3.3V I2C signal
Bi-directional
**Notice:
MOSFET BSS138 method cannot be used because:
- C-stray is larger (20pF-50pF) than IC (3.3-6pF) (Section 5.5 of IC)
- IC support higher open-drain speed (max 2 Mbps)(Section 1 of IC)
- IC use for I2C applications
- If BSS138 is used, need to include pull up resistor while IC has internal resistor
Routing: PCB trace need to be short otherwise it will cause ringing and oscilation

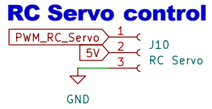
Telemetry 1



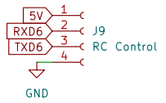
**Notice
1.Support extracting data through UART to come to a USB-to-UART device and transfer
it to a computer/laptop
2.Transmit data using RF module



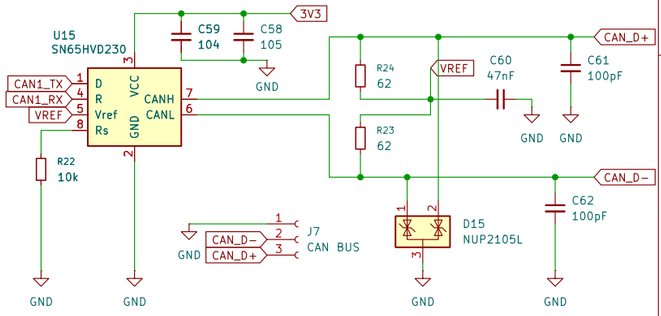
RC Servo control



C4001 24Ghz connector



CAN BUS



**Notice:
-TVS Diode absorbs energy surge to protect CAN_H and CAN_L
-Caps 100pF are placed to make a high frequency EMI filter
-Split termination need a 47nF combining with resistors of head and tail to make a LPF

Sheet: /Controller/
File: MCU-Sensors.kicad_sch

Title: Controller

Size: A3	Date: 2026-08-27	Rev:
KiCad E.D.A. 10.0.2		Id: 4/4